High Dynamic Range (HDR) Imaging

Chao-Tsung Huang

National Tsing Hua University

Department of Electrical Engineering

Wide dynamic range in real world

Unit of Luminous Intensity: Candela (cd)

1 cd = 

1 lumen = 1 cd·sr
(luminous flux; power)

1 lux = 1 lumen/m²
(illuminance)

over 10^9x!

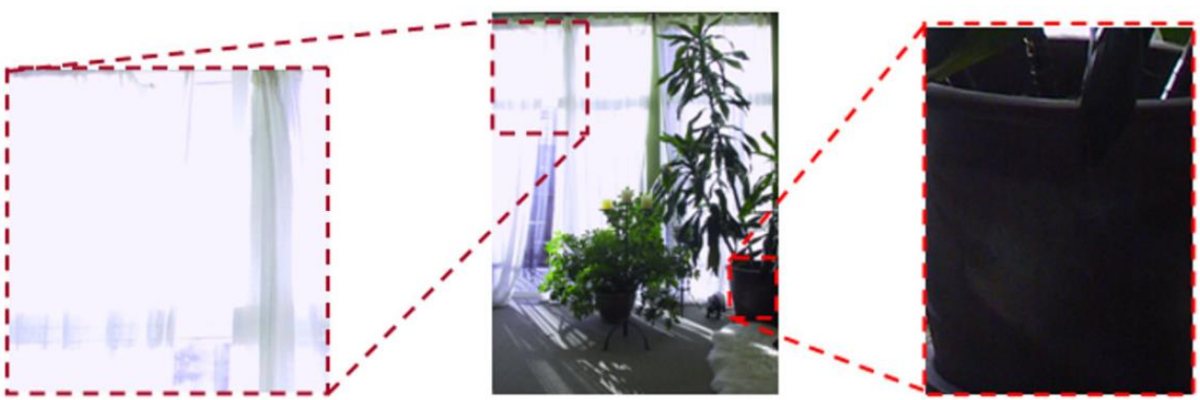
Examples	
Illuminance	**Surfaces illuminated by:**
10^{-4} lux	Moonless, overcast night sky (starlight)[2]
0.002 lux	Moonless clear night sky with airglow[2]
0.27 – 1.0 lux	Full moon on a clear night[2][3]
3.4 lux	Dark limit of civil twilight under a clear sky[4]
50 lux	Family living room lights (Australia, 1998)[5]
80 lux	Office building hallway/toilet lighting[6][7]
100 lux	Very dark overcast day[2]
320 – 500 lux	Office lighting[8][9][10]
400 lux	Sunrise or sunset on a clear day.
1000 lux	Overcast day;[2] typical TV studio lighting
10 000 – 25 000 lux	Full daylight (not direct sun)[2]
32 000 – 130 000 lux	Direct sunlight

Ref: *Wikipedia*

Wide dynamic range in real world

1 | 1,500 | 25,000 | 400,000 | 2,000,000

Relative brightness of different scenes



Over-exposure

Under-exposure

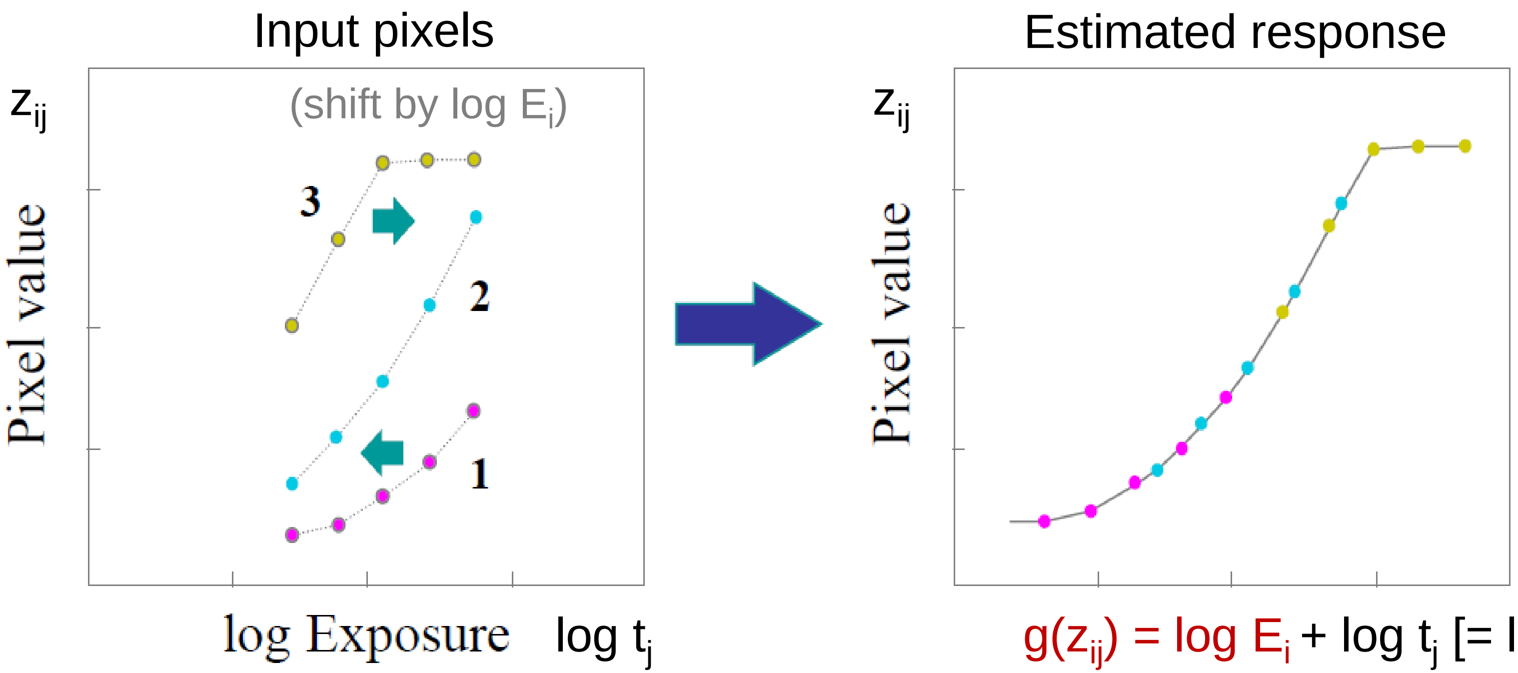
The dynamic range of common sensors is only about 10~14-bit ($10^{3\sim4}$x)

HDR Imaging (Chap 10.2)

- Composite shots with different exposures to include wider dynamic range

- Commonly in three steps:
  1. Estimate radiometric response function (e.g. gamma curve)
  2. Estimate (composite) the radiance map (like a super sensor)
  3. Tone mapping (for low dynamic range display)

1. Estimate radiometric response function



$g(z_{ij}) = \log E_i + \log t_j \; [= \log(E_i \, t_j)]$

z_{ij}: pixel value at position i in j-th image (known)
t_j: Exposure time for j-th image (known)
E_i: Radiance at position i (unknown)
g(k): Inverse response function (unknown, k=0~255)

Ref [1]: Debevec and Malik, "Recovering High Dynamic Range Radiance Maps from Photographs," SIGGRAPH '97. (website available)

Modeled in least squares problem

$$E = \sum_i \sum_j \underbrace{w(z_{i,j})}_{\text{Reliability weighting}} \underbrace{[g(z_{i,j}) - \log E_i - \log t_j]^2}_{\text{Curve fitting error}} + \underbrace{\lambda \sum_k w(k) g''(k)^2}_{\text{Regularization (second-order smoothness)}}$$

Reliability weighting

Curve fitting error

Regularization (second-order smoothness)

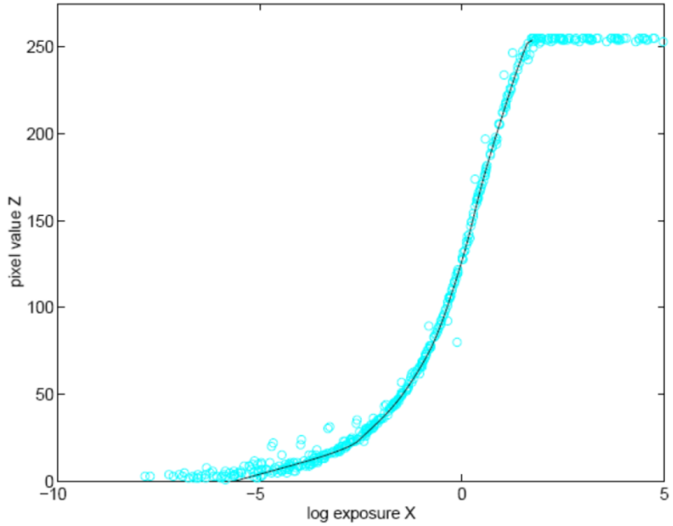
$$w(z) = \begin{cases} z - z_{\min} & z \le (z_{\min} + z_{\max})/2 \\ z_{\max} - z & z > (z_{\min} + z_{\max})/2 \end{cases}$$

$$g''(k) = g(k-1) - 2g(k) + g(k+1)$$

Solve using pseudo-inverse

Can be exactly solved using linear algebra

(21-line MATLAB code available on the last page in [1]!)



Quick guide:

Let $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{g} \\ \log \boldsymbol{E} \end{bmatrix}$.

Formulate $\mathrm{E} = \| \boldsymbol{AX} - \boldsymbol{b} \|^2$.

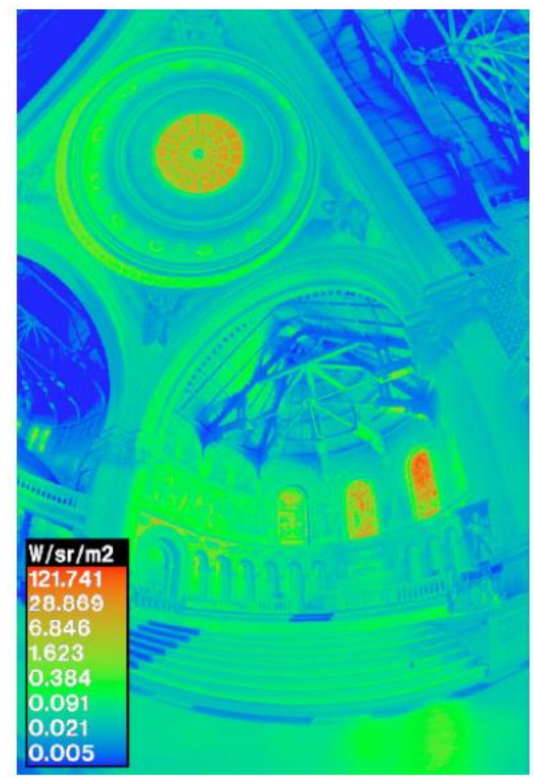
Solution: $\boldsymbol{X} = \boldsymbol{A}^{\dagger}\boldsymbol{b}$.

(pseudo-inverse by SVD)

2. Estimate radiance map

- Robust estimation with more emphasizing pixels values w/o saturation (less noise)

$$\log E_i = \frac{\sum_j w(z_{ij})[g(z_{ij}) - \log t_j]}{\sum_j w(z_{ij})} \qquad w(z) = g(z)/g'(z)$$

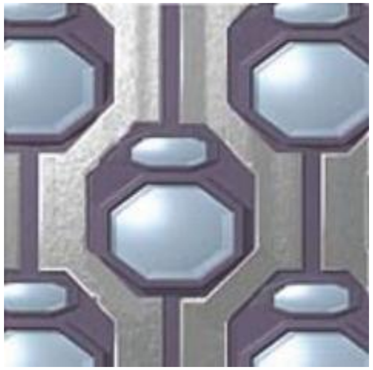


Misalignment problem

Composite directly

Composite w/ motion compensation

Physical approach to ease the first two steps



- Novel sensor configuration
  – e.g. Fuji SuperCCD
- Fast burst shots
  – e.g. Sony IMX 135 sensor
  – (two exposure shots for each frame)

Left: Without 'HDR movie' function

Comparison image

(13.13 effective megapixels(*3))

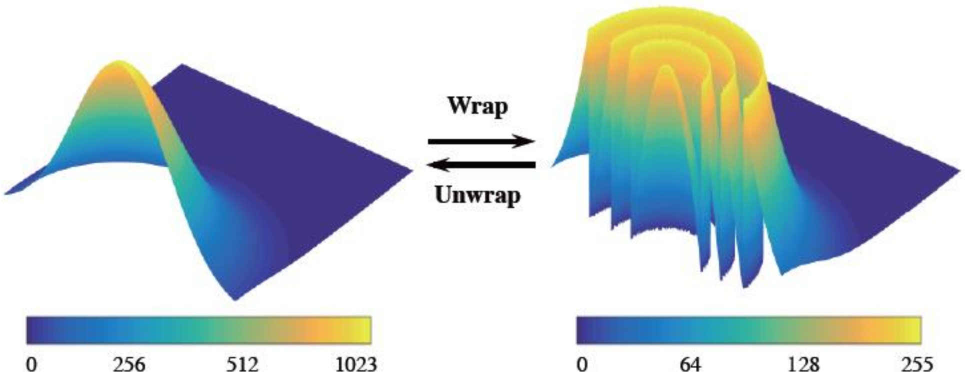
Right: With 'HDR movie' function

'IU135F3-Z'

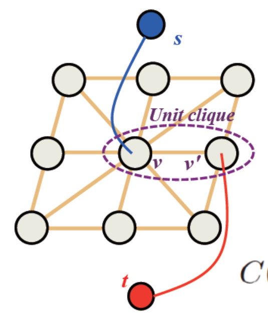
(13.13 effective megapixels(*3))

Modulo Camera

$$I_m(x,y) = \mathrm{mod}(I(x,y), 2^N)$$ (let sensor pixels never saturated)



Software Algorithm

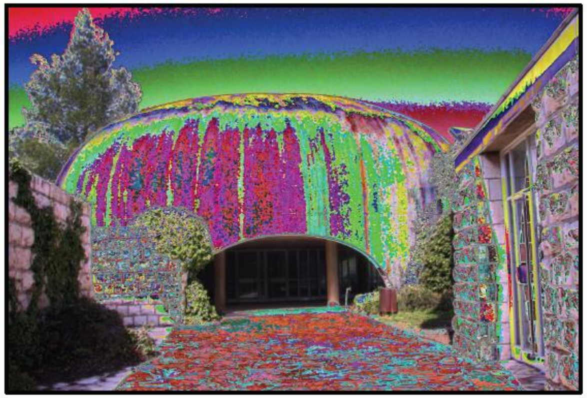


Markov Random Field
with 8 neighbors
Optimization with Graph Cuts

$$C(k|I_m) = \sum_{(i,j)\in\mathbb{G}} V(|\hat{I}_i - \hat{I}_j|)$$

Intensity camera

Modulo camera



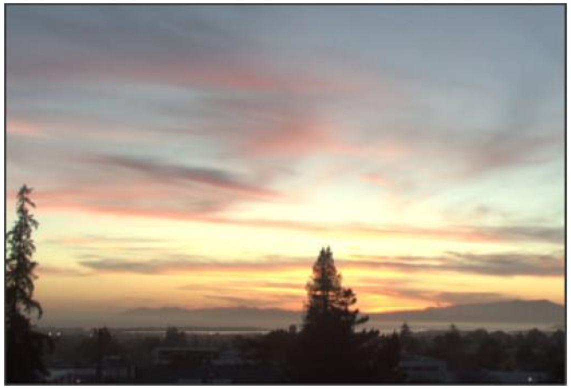
Recovered (tone mapped)

Ref [2]: H. Zhao, et. al., “Unbounded High Dynamic Range Photography using a Modulo Camera,” ICCP ’15 (runner-up). (website available)

3. Tone mapping

- Global tone mapping

Linear mapping



Additional gamma for RGB (less colorful)

Additional gamma for L* only

- Local tone mapping with linear filter
- Local tone mapping with bilateral filter

Local tone mapping

- Reduce the contrast locally as follows
  - Perform on luminance (e.g. L*) only

Build log luminance map
$$H(x,y) = \log L(x,y)$$

Build a base layer
$$H_{\mathrm{L}}(x,y) = B(x,y) * H(x,y)$$
(lowpass filter)

Build a detail layer
$$H_{\mathrm{H}}(x,y) = H(x,y) - H_{\mathrm{L}}(x,y)$$

Reduce contrast on base layer
$$H'_{\mathrm{L}}(x,y) = s\, H_{\mathrm{L}}(x,y)$$
(s usually < 1)

Derive new log luminance map
$$I(x,y) = H'_{\mathrm{L}}(x,y) + H_{\mathrm{H}}(x,y)$$

Use linear (Gaussian) filter for B(x,y)

base layer

detail layer

chrominance

Halo effect from detail layer

Tone-mapped result

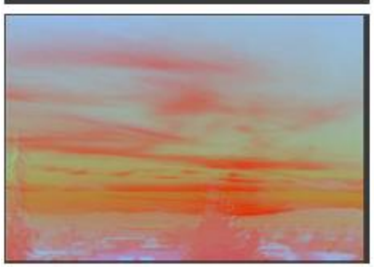
Use bilateral filter for B(x,y)

Edge-preserved filtering

base layer

detail layer

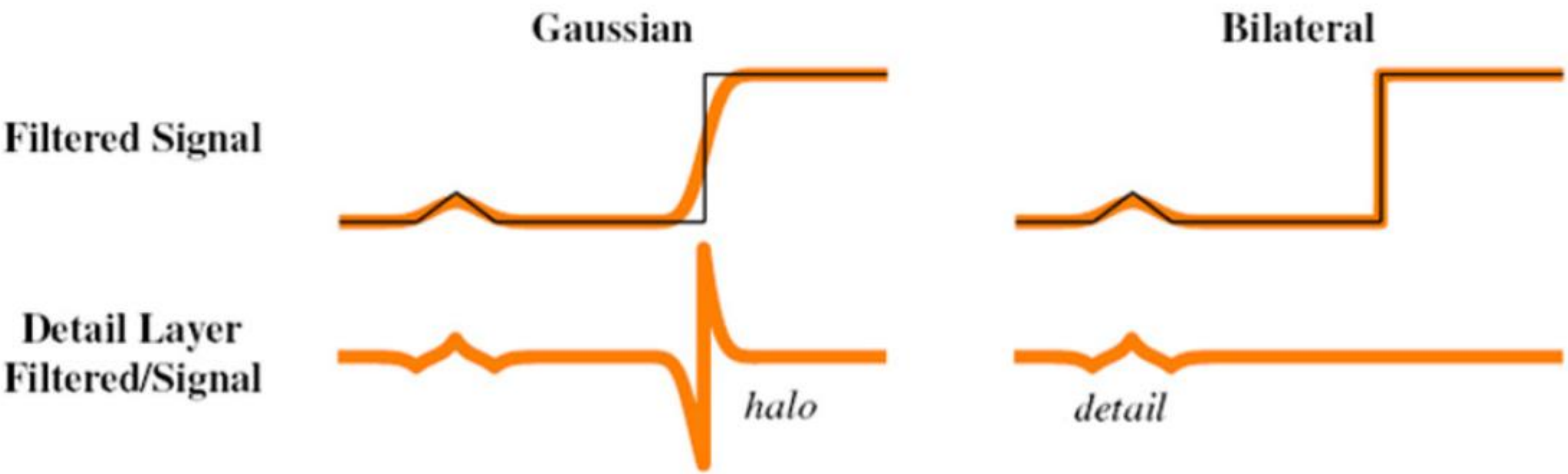
chrominance



Tone-mapped result

Gaussian vs. Bilateral filtering

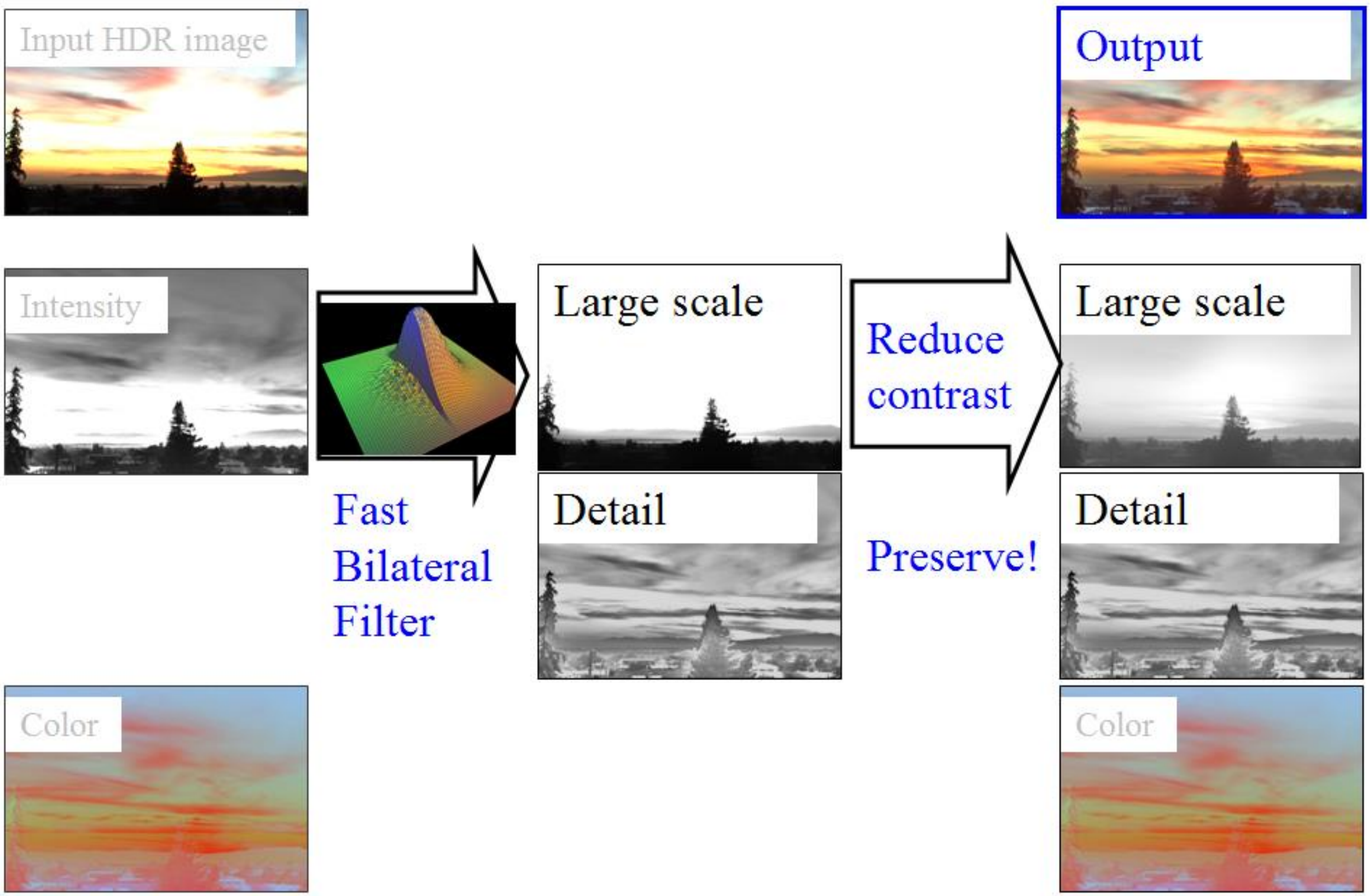
$$g(i,j) = \frac{\sum_{k,l} f(k,l)w(i,j,k,l)}{\sum_{k,l} w(i,j,k,l)}$$



$$w(i,j,k,l) = \exp\left(-\frac{(i-k)^2+(j-l)^2}{2\sigma_d^2}\right)$$

$$w(i,j,k,l) = \exp\left(-\frac{(i-k)^2+(j-l)^2}{2\sigma_d^2} - \frac{\|f(i,j)-f(k,l)\|^2}{2\sigma_r^2}\right)$$

Local tone mapping using bilateral filter



Ref [2]: Durand and Dorsey, “Fast Bilateral Filtering for the Display of High-Dynamic-Range Images,” SIGGRAPH ‘02. (website available)

Comparison

Global tone mapping

Local tone mapping using linear filter

Local tone mapping using bilateral filter